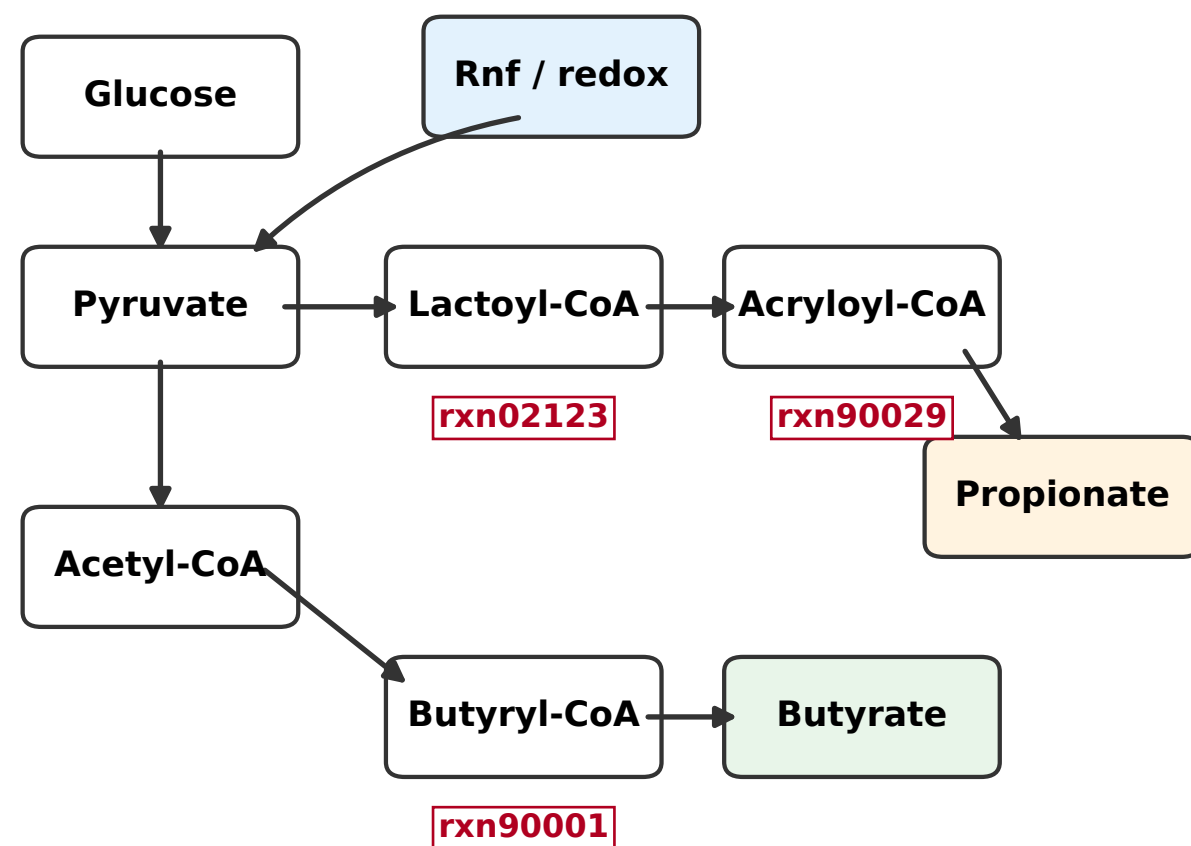
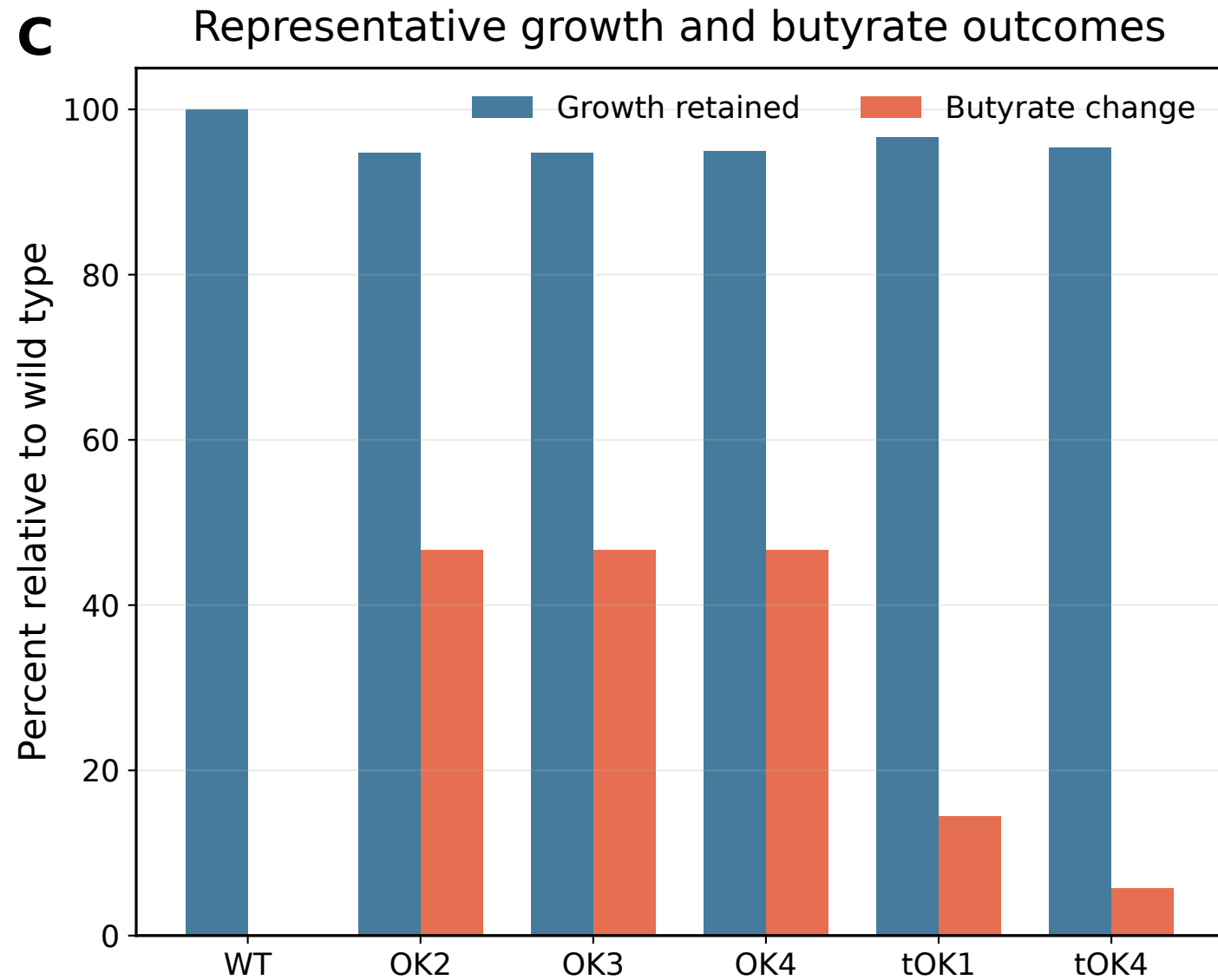


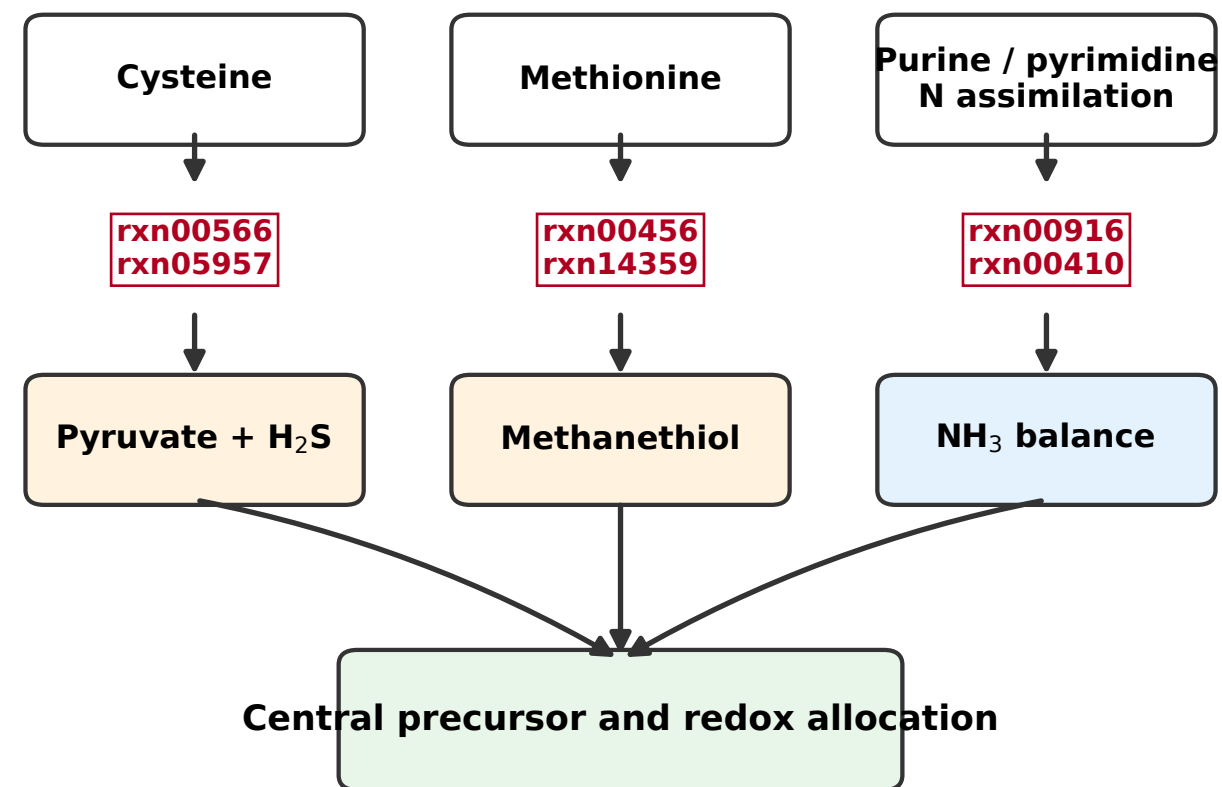
A Pyruvate and SCFA control architecture



Targets redirect flux between propionate and butyrate branches
Representative growth and butyrate outcomes



B Nitrogen and sulfur control architecture



Sulfur and nitrogen targets expose collateral product trade-offs

D Collateral product trade-offs

